

Intro to Computational Intelligence: Fall 2025

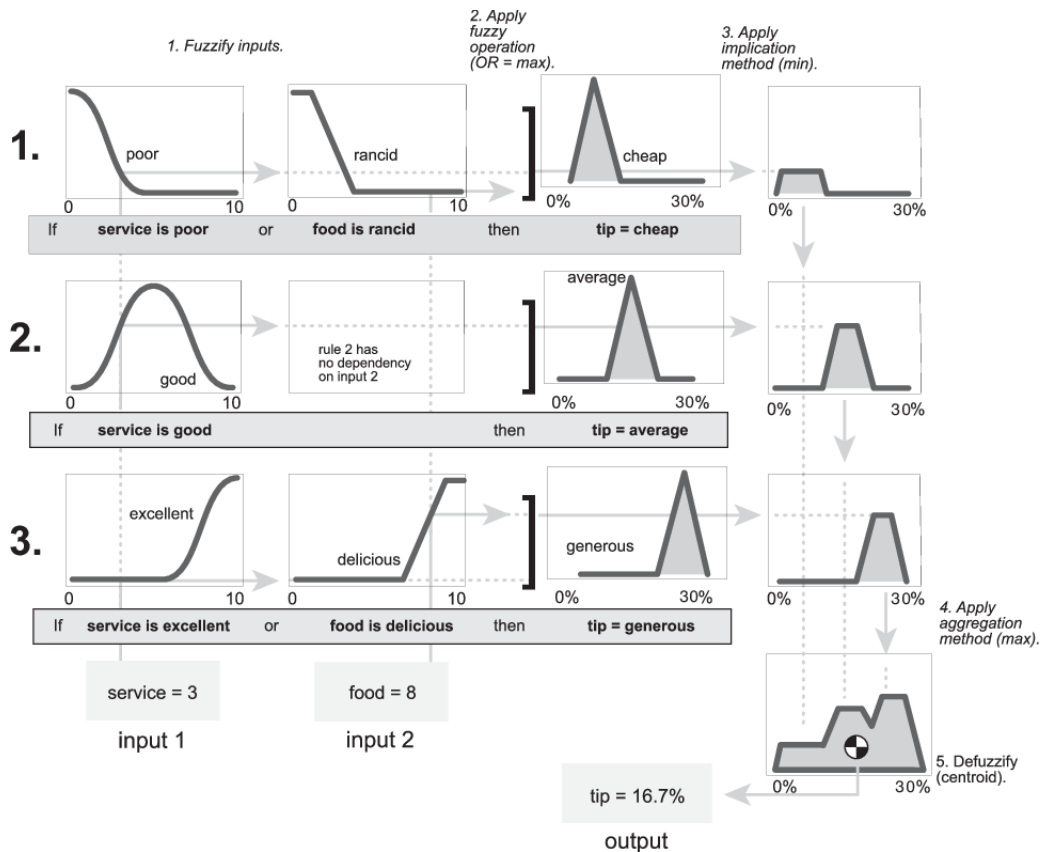
Project 2

Fuzzy Logic

Project Description

Implement a Mamdani style fuzzy inference system (FIS)

- From your book, but it's also graphically depicted at <https://www.mathworks.com/help/fuzzy/types-of-fuzzy-inference-systems.html> w.r.t. real-valued inputs



Your code needs to support:

- any number of rules
- multiple antecedents with AND (not OR)
- support for at least Gaussian and trapezoidal membership functions
- single consequent (you don't need to have rules with multiple THEN outputs)
- rules with different consequent domains (e.g., different linguistic variables/domains)
- support for different rule aggregation operators (e.g., max, sum)
- support for centroid defuzzification operator

Notes:

- you cannot use a “fuzzy library”, e.g., scikit-fuzzy
- you must code this up “from scratch”
 - You CAN do it all yourself (I encourage you)
 - You CAN use an LLM, but you better make sure what it writes is correct ☺

After your code is done, you must:

1. Validate your FIS
 - Sanity check / numeric example
 - Work it out by hand and make sure your program gives you the correct answer
 - Do for 2-3 examples (is 1 enough...?)
2. Pick a classification problem
 - E.g., IRIS dataset
 - https://scikit-learn.org/stable/auto_examples/datasets/plot_iris_dataset.html
 - fyi - there are three classes, four inputs
 - Questions (this is the “what you are actually doing, not just coding ☺”)
 - what should the linguistic variables and associated term sets be?
 - what membership functions should you use?
 - what should the consequence domain be?
 - how many and what rules?
 - what operations (e.g., antecedent combining, aggregation)?
 - what classification accuracy can/should you achieve?
 - what is a good train/val/test set?
 - do the rules / sets / etc. make sense? (explainable)
 - beyond classification, rule-base bring any other “value”?
 - etc.
 - Or pick something else (personal interest, graduate research, etc.)
3. Pick a non-classification problem
 - Regression problem, e.g., the tipping problem
 - https://pythonhosted.org/scikit-fuzzy/auto_examples/plot_tipping_problem_newapi.html
 - Similar questions to the above!
 - Control problem, e.g., the inverted pendulum
 - [https://en.wikipedia.org/wiki/Inverted_pendulum#:~:text=keep%20it%20balanced,-,The%20inverted%20pendulum%20is%20a%20classic%20problem%20in%20dynamics%20and,genetic%20algorithms%2C%20etc.\).](https://en.wikipedia.org/wiki/Inverted_pendulum#:~:text=keep%20it%20balanced,-,The%20inverted%20pendulum%20is%20a%20classic%20problem%20in%20dynamics%20and,genetic%20algorithms%2C%20etc.).)
 - Or pick something else (personal interest, graduate research, etc.)

Analysis:

- Like always, I am not looking for you to simply code this up. I want to know how you did it, what you tried, what you varied, etc. Want to know what you learned, what worked, did not work, etc. Why do you think it worked, did not work? Vary the rules. Vary the membership functions. Vary whatever in the system and learn about what we talked about in class !!!! This is your chance to learn about “where the rubber meets the road”!

Extra Credit: (10% each)

- **Mamdani FIS for real-valued singleton inputs**

- What to turn in?
 - Your code
 - Verify that it works
 - Compare your result to the 'regular' Mamdani FIS

- **Zadeh's discrete FIS for fuzzy-valued inputs**

- Your code must support:
 - any number of rules
 - modus ponens
 - {Lukasiewicz, corr min, corr product} implication
 - up to 3 antecedents with AND (not OR)
 - single consequent (you don't need to have rules with multiple THEN outputs)
 - rules with different consequent domains (e.g., different linguistic variables/domains)
 - support for different rule aggregation operators (e.g., max, sum)
- What to turn in?
 - Your code
 - Verify that it works
 - Pass numeric examples from your book to verify it works
 - E.g., pass the rule in, not rule, sort of matching rule, etc.

Report Requirements

Your final report must follow these guidelines:

1. Cover Slide: Title, name, student ID, date.
2. Technical Description (5–10 pages).
3. Validation, Experiments, Results (10–20 pages).
4. Summary and Future Work (3–5 pages).

Code Submission

Submit your code as a single ZIP (under 50MB, no data).

Rules and AI/GenAI/LLM Use Policy

You must work individually. Collaboration is not allowed. The ONLY exception is AI. You **may** use AI tools such as ChatGPT for generating code and polishing your report (e.g., help with grammar, formatting, organization). Be careful, you are responsible for the code you use. If you use an AI tool and its wrong, that is on you. You must also be the one to write the initial report. Meaning the ideas and findings and summaries are yours. AI cannot be used for this. When in doubt, consult your TA or instructor.